Year 9 · Chapter 10 · Surface area and volume: answers (teacher copy)

10.1 Metric units (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: 8 × 1000 = 8 000 m
2. Ex 1b: 250 ÷ 100 = 2.5 m
3. Ex 1c: 3.5 × 1000 = 3 500 g
4. Ex 1d: 4500 ÷ 1000 = 4.5 L
5. Ex 2a: 3.5 × 60 = 210 min
6. Ex 2b: 150 ÷ 60 = 2.5 min
7. Ex 2c: 4.2 × 1000 = 4 200 MB
8. Ex 2d: 2 500 000 ÷ 1 000 000 = 2.5 m

Easy

1. 6 000 m
2. 3 km
3. 400 cm
4. 7 m
5. 90 mm
6. 12 cm
7. 2 500 m
8. 0.45 m
9. 5 000 g
10. 2 kg
11. 3 000 kg
12. 6.5 t
13. 800 g
14. 0.75 kg
15. 4 g
16. 1 200 kg
17. 7 000 mL
18. 3 L
19. 2 000 L
20. 4.5 kL
21. 1 500 mL
22. 0.25 L
23. 600 L
24. 0.08 L
25. 400 000 cm
26. 3 000 mm
27. 0.6 km
28. 5 m
29. 2 000 000 mg
30. 1 000 000 mL
31. 50 000 cm
32. 0.08 m

Medium

1. 300 s
2. 240 min
3. 3 min
4. 150 min
5. 1.5 h
6. 72 h
7. 4 s
8. 90 s
9. 3 000 MB
10. 4.5 GB
11. 2 400 GB
12. 34 kB
13. 2 600 000 m
14. 700 ms
15. 3 m
16. 5 000 µs
17. 80 videos
18. 2 h 15 min

Mixed practice

1. 750 m (10.1)
2. 40 cm² (Year 8)
3. 1.2 m (= 120 cm) (10.1)
4. 3.5 (Year 8)
5. 7 200 s (10.1)
6. 28 m (Year 8)
7. 5 cups (10.1)
8. 2 500 MB (10.1)

10.2 Precision and absolute error (booklet p. 5)

WE DO examples (teacher)

1. Ex 1a: precision 1 mm, absolute error 0.5 mm
2. Ex 1b: precision 50 mL, absolute error 25 mL
3. Ex 1c: precision 1 m, absolute error 0.5 m
4. Ex 1d: precision 0.1 kg, absolute error 0.05 kg
5. Ex 2a: precision 0.01 m, absolute error 0.005 m
6. Ex 2b: 2.445 m to 2.455 m
7. Ex 2c: 161.5 cm to 162.5 cm
8. Ex 2d: 625 mL to 675 mL

Easy

1. 1 m
2. 0.1 cm
3. 0.01 kg
4. 1 mm
5. 0.1 L
6. 0.01 t
7. 1 g
8. 0.1 s
9. 0.5 m
10. 0.05 kg
11. 0.5 mL
12. 0.005 m
13. 0.5 cm
14. 0.05 L
15. 0.005 s
16. 0.5 g
17. precision 0.1 kg, absolute error 0.05 kg
18. precision 20 mL, absolute error 10 mL
19. precision 2 °C, absolute error 1 °C
20. precision 0.5 m, absolute error 0.25 m

Medium

1. 0.005 kg
2. 0.05 m
3. 0.05 L
4. 0.0005 km
5. 0.005 g
6. 0.005 m
7. 0.05 cm
8. 0.0005 t
9. 23.5 m to 24.5 m
10. 6.35 cm to 6.45 cm
11. 152.5 g to 153.5 g
12. 3.245 kg to 3.255 kg
13. 9.75 s to 9.85 s
14. 71.5 kg to 72.5 kg
15. 1.55 L to 1.65 L
16. 0.445 m to 0.455 m
17. 245 g to 255 g
18. No: the limits are 12.45 m to 12.55 m, and 12.56 m is longer than 12.55 m

Mixed practice

1. 0.01 m (10.2)
2. 3 200 m (10.1)
3. 0.5 mm (10.2)
4. 3 (Year 8)
5. 17.5 cm to 18.5 cm (10.2)
6. 0.75 L (10.1)
7. 10 g (10.2)
8. 0.05 L (10.2)

10.3 Percentage error (booklet p. 7)

WE DO examples (teacher)

1. Ex 1a: 0.5 ÷ 20 × 100 = 2.5%
2. Ex 1b: 0.5 ÷ 4 × 100 = 12.5%
3. Ex 1c: 0.05 ÷ 2.5 × 100 = 2%
4. Ex 1d: 0.5 ÷ 50 × 100 = 1%
5. Ex 2a: 0.5 m; 0.5 ÷ 24 × 100 = 2.08%
6. Ex 2b: 0.005 ÷ 3.25 × 100 = 0.15%
7. Ex 2c: 12.5 ÷ 375 × 100 = 3.33%
8. Ex 2d: b: it has the smallest percentage error (0.15%)

Easy

1. 5%
2. 2%
3. 1%
4. 12.5%
5. 2.5%
6. 0.5%
7. 10%
8. 1.25%
9. 2%
10. 10%
11. 3.125%
12. 0.4%
13. 25%
14. 12.5%
15. 0.08%
16. 6.25%
17. 7.1%
18. 3.3%
19. 5.6%
20. 1.7%
21. 1.4%
22. 8.3%
23. 1.1%
24. 2.2%

Medium

1. 2.08%
2. 1.39%
3. 0.04%
4. 0.29%
5. 6.25%
6. 0.21%
7. 1.04%
8. 0.53%
9. 1.43%
10. 16.67%
11. 8.33%
12. 2.08%
13. height: 0.30% is smaller than 3.33%
14. 0.40%

Mixed practice

1. 6.25% (10.3)
2. 5.25 kg to 5.35 kg (10.2)
3. 2.5% (10.3)
4. 4.5 kg (10.1)
5. 10% (10.3)
6. 100 m (0.5%, not 5%) (10.3)
7. 30 cm² (Year 8)
8. 2% (10.3)

10.4 Perimeter and area of composite shapes (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: missing sides 6 m and 5 m; P = 10 + 4 + 6 + 5 + 4 + 9 = 38 m
2. Ex 1b: 10 × 4 + 4 × 5 = 40 + 20 = 60 m²
3. Ex 1c: 8 × 5 + ½ × 8 × 3 = 40 + 12 = 52 cm²
4. Ex 1d: 10 × 6 − 3 × 2 = 60 − 6 = 54 m²
5. Ex 2a: the steps add to 12 m across and 9 m up: P = 2 × (12 + 9) = 42 m
6. Ex 2b: 12 × 3 + 8 × 3 + 4 × 3 = 72 m²
7. Ex 2c: 7.5 × 4 + ½ × 7.5 × 2.4 = 30 + 9 = 39 cm²
8. Ex 2d: 12 × 8 − ½ × 5 × 4 = 96 − 10 = 86 m²

Easy

1. 30 cm
2. 44 m
3. 48 m
4. 46 m
5. 36 cm²
6. 50 m²
7. 33 cm²
8. 55 m²
9. 48 m²
10. 39 cm²
11. 42 m²
12. 45 cm²

Medium

1. 36 cm
2. 37 m
3. 28 m
4. 40 cm
5. 42 m²
6. 66 cm²
7. 85 m²
8. 64 cm²

Mixed practice

1. 34 m (10.4)
2. 1.25% (10.3)
3. 90 m² (10.4)
4. 240 cm (10.1)
5. 18 m² (10.4)
6. 7.15 m to 7.25 m (10.2)
7. 18 m² (10.4)
8. 28 cm (10.4)

10.5 Areas of quadrilaterals (booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: A = bh = 12 × 4 = 48 cm²
2. Ex 1b: A = ½(a + b)h = ½ × (14 + 8) × 6 = 66 cm²
3. Ex 1c: A = ½xy = ½ × 10 × 6 = 30 m²
4. Ex 1d: A = ½xy = ½ × 8 × 12 = 48 cm²
5. Ex 2a: the 5 cm side is not needed: A = 12 × 4 = 48 cm²
6. Ex 2b: ½ × (7.5 + 4.5) × 3.2 = 19.2 m²
7. Ex 2c: 54 = 9 × h, so h = 54 ÷ 9 = 6 cm
8. Ex 2d: ½ × 1.2 × 0.8 = 0.48 m²

Easy

1. 40 cm²
2. 48 m²
3. 105 cm²
4. 45 m²
5. 32 cm²
6. 50 m²
7. 42 cm²
8. 36 m²
9. 24 cm²
10. 30 m²
11. 42 cm²
12. 72 m²

Medium

1. 36 m²
2. 40 cm²
3. 96 cm²
4. 40.5 m²
5. 8 cm
6. 6 m
7. 5 cm
8. 50 cm²

Mixed practice

1. 21 m² (10.5)
2. 26 cm (10.4)
3. 25 cm² (10.5)
4. 2% (10.3)
5. 18 m² (10.5)
6. 600 m (10.1)
7. 35 cm² (10.5)
8. 5 m (10.5)

10.6 Circumference and arc length (booklet p. 13)

WE DO examples (teacher)

1. Ex 1a: C = 2 × π × 5 = 31.4 cm
2. Ex 1b: C = π × 12 = 37.7 m
3. Ex 1c: P = ½ × π × 10 + 10 = 25.7 cm
4. Ex 1d: P = ¼ × 2 × π × 6 + 6 + 6 = 21.4 m
5. Ex 2a: l = 60/360 × 2 × π × 9 = 9.4 cm
6. Ex 2b: P = 9.42… + 9 + 9 = 27.4 cm
7. Ex 2c: P = 10 + 6 + 10 + ½ × π × 6 = 35.4 m (not the dashed side)
8. Ex 2d: 50 = 2 × π × r, so r = 50 ÷ (2 × π) = 8.0 cm

Easy

1. 18.8 cm
2. 44.0 m
3. 62.8 cm
4. 15.7 m
5. 50.3 mm
6. 9.4 km
7. 125.7 cm
8. 40.8 m
9. 25.1 cm
10. 47.1 m
11. 94.2 mm
12. 6.3 m
13. 28.3 cm
14. 157.1 cm
15. 4.4 km
16. 78.5 m
17. 20.6 cm
18. 35.7 m
19. 36.0 m
20. 14.3 cm

Medium

1. 24.6 cm
2. 22.3 m
3. 26.8 cm
4. 46.2 m
5. 28.6 cm
6. 25.4 m
7. 30.3 m
8. 25.9 cm

Mixed practice

1. 56.5 cm (10.6)
2. 23 m (10.4)
3. 201.1 m (10.6)
4. 3.45 m (10.1)
5. 3.0 m (10.6)
6. 15 m² (10.5)
7. 6.3 m (10.6)
8. 51.4 cm (10.6)

10.7 Areas of circles and sectors (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: A = π × 5² = 78.5 cm²
2. Ex 1b: r = 12 ÷ 2 = 6 m; A = π × 6² = 113.1 m²
3. Ex 1c: r = 10 ÷ 2 = 5 cm; A = ½ × π × 5² = 39.3 cm²
4. Ex 1d: A = ¼ × π × 6² = 28.3 m²
5. Ex 2a: A = $\frac{120}{360}$ × π × 9² = 84.8 cm²
6. Ex 2b: 10 × 6 + ½ × π × 3² = 60 + 14.1 = 74.1 m²
7. Ex 2c: π × 10² − π × 6² = 314.2 − 113.1 = 201.1 cm²
8. Ex 2d: the circle's diameter is 8 cm: 8 × 8 − π × 4² = 64 − 50.3 = 13.7 cm²

Easy

1. 28.3 cm²
2. 153.9 m²
3. 314.2 mm²
4. 50.3 cm²
5. 19.6 m²
6. 254.5 cm²
7. 7.1 km²
8. 452.4 m²
9. 201.1 m²
10. 380.1 mm²
11. 3.1 m²
12. 706.9 cm²
13. 38.5 cm²
14. 530.9 m²
15. 1 963.5 cm²
16. 1.1 km²
17. 77.0 cm²
18. 50.3 m²
19. 157.1 m²
20. 95.0 cm²

Medium

1. 18.8 cm²
2. 37.7 m²
3. 56.5 cm²
4. 83.8 m²
5. 122.5 m²
6. 137.1 cm²
7. 60.6 m²
8. 51.7 cm²

Mixed practice

1. 12.6 cm² (10.7)
2. 31.4 m (10.6)
3. 63.6 m² (10.7)
4. 20 m² (10.5)
5. 88.4 cm² (r = 15 cm; each slice is a 45° sector) (10.7)
6. 12.5% (10.3)
7. 7 cm ($r^2 = 153.9 \div \pi \approx 49$) (10.7)
8. 87.4 m² (10.7)

10.8 Surface area of a prism (booklet p. 17)

WE DO examples (teacher)

1. Ex 1a: SA = $6 \times 5^2 = 6 \times 25 = 150$ m²
2. Ex 1b: SA = $2 \times (8 \times 5 + 8 \times 4 + 5 \times 4) = 2 \times 92 = 184$ cm²
3. Ex 1c: 5 faces: 2 triangles (the ends) and 3 rectangles
4. Ex 1d: no top: $184 - 8 \times 5 = 184 - 40 = 144$ cm²
5. Ex 2a: $2 \times (\frac{1}{2} \times 8 \times 6) + (8 + 6 + 10) \times 12 = 48 + 288 = 336$ cm²
6. Ex 2b: the right angle is between 5 and 12: $2 \times (\frac{1}{2} \times 5 \times 12) + (5 + 12 + 13) \times 10 = 60 + 300 = 360$ cm²
7. Ex 2c: inside edges $6 - 2 = 4$ and $5 - 2 = 3$; end: area $6 \times 2 + 2 \times 3 = 18$, perimeter $2 \times (6 + 5) = 22$; SA = $2 \times 18 + 22 \times 8 = 36 + 176 = 212$ cm²
8. Ex 2d: walls $2 \times (5 \times 2.5 + 4 \times 2.5) = 45$, ceiling $5 \times 4 = 20$: $45 + 20 = 65$ m²

Easy

1. 24 cm²
2. 96 m²
3. 600 mm²
4. 294 cm²
5. 13.5 m²
6. 486 cm²
7. 54 m²
8. 2 400 mm²
9. 108 cm²
10. 160 m²
11. 142 cm²
12. 228 mm²
13. 96 cm²
14. 180 cm²
15. 132 m²
16. 196 cm²

Medium

1. 132 cm²
2. 300 m²
3. 520 cm²
4. 396 m²
5. 2 200 cm²
6. 9 000 cm²

Mixed practice

1. 384 cm² (10.8)
2. 96 cm² ($\frac{1}{2} \times 12 \times 16$) (10.4)
3. 248 cm² (10.8)
4. 28 m² (10.5)
5. 3 m ($54 \div 6 = 9 = 3^2$) (10.8)
6. 180 cm (10.1)
7. 256 cm² (10.8)
8. 16 m² (10.8)

10.9 Surface area of a cylinder (booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: $A = \pi \times 3^2 = 28.3$ cm²
2. Ex 1b: $A = 2 \times \pi \times 3 \times 8 = 150.8$ cm²
3. Ex 1c: $SA = 2 \times \pi \times 3^2 + 2 \times \pi \times 3 \times 8 = 207.3$ cm²
4. Ex 1d: r = 10 ÷ 2 = 5 m; $SA = 2 \times \pi \times 5^2 + 2 \times \pi \times 5 \times 4 = 282.7$ m²
5. Ex 2a: one end only: $SA = \pi \times 4^2 + 2 \times \pi \times 4 \times 10 = 301.6$ cm²
6. Ex 2b: r = 14 ÷ 2 = 7 cm; $SA = 2 \times \pi \times 7^2 + 2 \times \pi \times 7 \times 20 = 1\,187.5$ cm²
7. Ex 2c: the label is the curved surface: $2 \times \pi \times 3.5 \times 11 = 241.9$ cm²
8. Ex 2d: no ends: $2 \times \pi \times 2 \times 50 = 628.3$ cm²

Easy

1. 62.8 cm²
2. 188.5 m²
3. 56.5 m²
4. 131.9 mm²
5. 125.7 cm²
6. 754.0 cm²
7. 18.8 m²
8. 94.2 m²
9. 100.5 cm²
10. 251.3 m²
11. 326.7 cm²
12. 51.8 m²
13. 226.2 cm²
14. 534.1 m²
15. 942.5 cm²
16. 213.6 m²

Medium

1. 455.5 cm²
2. 175.9 m²
3. 377.0 cm²
4. 1 131.0 cm²
5. r = 2 ÷ 2 = 1 m; $SA = 2 \times \pi \times 1^2 + 2 \times \pi \times 1 \times 1.5 = 15.7$ m²; cost = 15.708… × $12 = $188.50 ($188.40 using 15.7 m²)
6. r = 6 ÷ 2 = 3 cm; one turn = $2 \times \pi \times 3 \times 25 = 471.2$ cm²; 10 turns = 4 712 cm²

Mixed practice

1. 219.9 cm² (10.9)
2. 113.1 cm² (10.7)
3. 603.2 cm² (10.9)
4. 76 cm² (10.8)
5. 3 063.1 cm² (10.9)
6. 11.45 cm to 11.55 cm (10.2)
7. 5 cm ($r = 314.2 \div (2 \times \pi \times 10)$) (10.9)
8. neither: $2 \times \pi \times 2 \times 6 = 2 \times \pi \times 3 \times 4 = 75.4$ cm² (10.9)

10.10 Volumes of prisms and cylinders

(booklet p. 21)

WE DO examples (teacher)

1. Ex 1a: $V = l \times w \times h = 8 \times 5 \times 4 = 160$ cm³
2. Ex 1b: $V = 5 \times 5 \times 5 = 5^3 = 125$ m³
3. Ex 1c: $V = Ah = 20 \times 6 = 120$ cm³
4. Ex 1d: $V = \pi r^2 h = \pi \times 3^2 \times 8 = 226.2$ cm³
5. Ex 2a: $A = \frac{1}{2} \times 8 \times 6 = 24$ cm²; $V = 24 \times 12 = 288$ cm³ (the 10 cm side is not needed)
6. Ex 2b: $A = 6 \times 2 + 2 \times 3 = 18$ cm²; $V = 18 \times 8 = 144$ cm³
7. Ex 2c: r = 10 ÷ 2 = 5 m; $V = \pi \times 5^2 \times 4 = 314.2$ m³
8. Ex 2d: $V = 50 \times 30 \times 40 = 60\,000$ cm³ = 60 000 mL = 60 L

Easy

1. 72 cm^3
2. 100 m^3
3. 27 cm^3
4. 200 mm^3
5. 480 cm^3
6. 216 m^3
7. 30 m^3
8. 1 000 cm^3
9. 105 cm^3
10. 63 m^3
11. 96 cm^3
12. 140 m^3
13. 62.8 cm^3
14. 235.6 m^3
15. 351.9 cm^3
16. 42.4 m^3

Medium

1. 300 m^3
2. 130 cm^3
3. 251.3 m^3
4. 180 cm^3
5. r = 1.5 m; $V = \pi \times 1.5^2 \times 2 = 14.137\ldots\ m^3 = 14\ 137$ L
6. $V = 25 \times 10 \times 1.5 = 375\ m^3 = 375\ 000$ L

Mixed practice

1. 300 cm^3 (10.10)
2. 50.3 cm^2 (10.7)
3. 80 m^3 (10.10)
4. 88.0 m^2 (10.9)
5. 282.7 mL (r = 3 cm; 1 cm^3 = 1 mL) (10.10)
6. 3 500 mL (10.1)
7. 5 cm (120 ÷ 24) (10.10)
8. 1 L (1 000 cm^3) (10.10)